

Antoine Lorient

APPLIED MACHINE LEARNING FOR HUMAN AND COMPUTER INTERACTION

3 rue de la Croix Faubin, 75011 Paris

☎ (+33) 7-66-73-95-93 | ✉ antoine.lorient@gmail.com | 📱 toinsson | 🌐 antinelorient

Skills

Data Modelling	Python, Pytorch, Keras, Cuda, MFlow, Pyro, Seaborn, Jupyter
Applied Research	Scientific Writing, Quantitative Analysis, Experiment Design, Ethics
Management	Lean, Agile, Scrum Master, Teaching, Tutoring

Research

CNRS - IRCAM - INRIA

Paris, France

POST-DOCTORAL RESEARCHER

Jan. 2020 - Sept. 2022

- Created a functioning prototype (Javascript, MaxMSP) for movement exploration and modeling, demoed at NIME 2022.
- Studied the perception of motor learning (Pytorch) in tasks involving complex and creative motions, published in PLOS ONE.

University of Glasgow

Glasgow, United Kingdom

RESEARCH ASSISTANT AND PHD CANDIDATE

Mar. 2015 - Sep. 2019

- Researched computational methods for the study of novel gestural interactions (thesis). Use of discriminative models to create virtual surfaces from RGB-D sensors (*paper*); optimisation of gameplay experiences for custom game controllers via user probabilistic modelling (*paper*); audio-based reinforcement learning for the exploration and characterisation of user physical capabilities.
- Contributions to the European project MoreGrasp: lead redaction of deliverable documents, implementation of visual feedback mechanisms on smartwatches, preparation of live demos and posters for review meetings, establishment of contacts with practitioners in the QEU hospital in Glasgow leading to workshops in-situ and the writing of a funding proposal targeting a local charity.
- Analysed visualisation techniques (*Jupyter notebooks*) for low-dimensional embeddings (t-SNE).
- Tutored for Artificial Intelligence and Data Fundamentals courses (undergraduate, Master levels).

Mobile Vinn Excellence Center

Stockholm, Sweden

RESEARCH ASSISTANT

Feb. 2014 - Oct. 2014

- Research on innovative designs for always-on in the background speech interactions (*paper*). Design of an experiment involving day-long recordings of participants' everyday life. Qualitative analysis and design workshop with trained Interaction designers.

Industry

Now & Net

Paris, France

PARTNER

2013 - 2015

- Co-founder of a startup providing a collective musical experience organised around DJ-ing.
- Developed of a POC recommender system (Neo4j)

Ericsson AB

Stockholm, Beijing, Seattle

MULTIPLE POSITIONS

2008 - 2013

- SCRUM master and Software Developer [2013/2014]: Acting as the "guardian of the process" to facilitate daily operations of a cross functional team (8 people). Animation of planning meetings, daily follow-up meetings and retrospective meetings.
- System engineer AT&T laboratory [summer 2012]: Feature integration into the customer network with laboratory and in-the-field experiments.
- Research engineer at Ericsson Research [2012]: Preparation of the 4G+ prototype for the Mobile World Congress (MWC).
- Software Developer 4G/LTE [2008/2011]: Development in C of algorithms from standard specifications (3GPP). Extension and support of an Erlang test framework used by 50 engineers, including training sessions in Stockholm and Beijing.

Education

University of Glasgow

Glasgow, United Kingdom

PHD IN COMPUTING SCIENCE

2019

Thesis title: A Computational Approach to Gestural Interactions of the Upper Limb on Planar Surfaces.

Royal Institute of Technology (KTH)

Stockholm, Sweden

MSC IN APPLIED PHYSICS

2009

Telecom Physique Strasbourg

Illkirch-Graffenstaden, France

MSC IN ENGINEERING AND PHYSICS

2008